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A Hybrid Legendre–Laguerre Operational Matrix Method for the Variable-Coefficient Fifth-Order KdV Equation in Non-Homogeneous Optical Media

Cenk Keşan*

Department of Mathematics, Faculty of Science, Dokuz Eylül University, 35390 İzmir, Turkey

* Correspondence: Cenk Keşan (cenk.kesan@deu.edu.tr)

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Abstract: Light trapping in microcavities and non-homogeneous optical media has become increasingly important for the development of quantum information processing, slow-light devices, integrated photonic circuits, and bio-inspired optical systems. However, conventional optical trapping models based on fixed geometric configurations or standard nonlinear Schrödinger formulations are generally unable to accurately represent the combined effects of high-order dispersion, spatially varying material properties, and structural asymmetry encountered in realistic optical environments. To address these limitations, a hybrid shifted Legendre–classical Laguerre operational matrix method was developed for the numerical solution of the variable-coefficient fifth-order Korteweg–de Vries (KdV) equation. The spatial domain, $x \in [0, L]$, is discretized using shifted Legendre polynomials to facilitate accurate enforcement of algebraic boundary conditions, whereas the semi-infinite temporal domain, $t \in [0, \infty)$, is represented by classical Laguerre polynomials that remain orthogonal with respect to the exponentially decaying weight function $w(t) = e^{-t}$. By combining spectral collocation with Chebyshev–Gauss–Lobatto nodes, the governing nonlinear partial differential equation was transformed into a square system of nonlinear algebraic equations, which was subsequently solved using a trust-region dogleg algorithm. Numerical results show that, under the variable coefficients, the pulse is not trapped: over $t \in [0, 5]$, its peak drifts from $x \approx 5$ toward $x \approx 8.9$ and its amplitude decays from ≈ 2.0 to ≈ 0.7 as it sheds dispersive radiation, so the monitored L^2 quantity decreases to about 0.18 of its initial value; the proposed scheme resolves this drift-and-decay evolution stably. A controlled comparison against a fourth-order finite-difference scheme under identical time integration shows that the spectral discretization reaches a given accuracy with roughly an order of magnitude fewer unknowns and is several orders of magnitude more accurate at equal degrees of freedom. The fixed-point iteration is convergent, with the Jacobian spectral radius below one. The proposed framework provides an efficient and mathematically rigorous spectral methodology for modeling variable-coefficient higher-order soliton dynamics in non-homogeneous media, and establishes a unified computational framework for such problems. Although the Laguerre basis is defined on $[0, \infty)$, all reported simulations are evaluated on the finite window $t \in [0, 5]$; the semi-infinite construction supplies a spectrally accurate temporal basis with the natural weight e^{-t}, and $[0, 5]$ covers the dynamics of interest.

Keywords: Hybrid operational matrix; Shifted Legendre polynomials; Classical Laguerre polynomials; Variable-coefficient fifth-order Korteweg–de Vries equation; Soliton drift and decay; Spectral collocation

1 Introduction

The phenomenon of trapping light within microscopic domains has emerged as a cornerstone of modern photonics, driving advancements in optical storage, quantum information processing, and slow-light switches [1, 2]. Traditional approaches rely primarily on static geometric modifications, such as engineered photonic crystal defects or metamaterial boundaries. While effective under rigid constraints, these mechanisms lack the dynamic adaptability required for real-time environmental perturbations in complex non-homogeneous optical media.

To overcome these limitations, the optical Kerr effect has been exploited to generate stable solitons [3–7]. While the nonlinear Schrödinger equation and the standard Korteweg–de Vries (KdV) equation are widely employed as classical integrable models [8, 9], they overlook high-order spatial dispersion [10, 11]. Fifth-order KdV equations address this by introducing $\partial^5 u/\partial x^5$ terms. Spectral methods achieve exponential convergence for smooth problems [12, 13] with far fewer degrees of freedom than the finite-difference time-domain method or the finite element method.

The central innovation of this study is a hybrid shifted Legendre–Laguerre operational matrix approach combining two orthogonal polynomial families into a single tensor-product framework, extending established operational-matrix spectral techniques [14–20]. This study is structured as follows: Section 2 develops the mathematical framework. Section 3 describes the numerical solution procedure. Section 4 presents results and validation. Section 5 concludes.

2 Mathematical Formulation

2.1 Variable-Coefficient Fifth-Order Korteweg–de Vries Equation

This study considers the initial–boundary value problem governed by the variable-coefficient fifth-order KdV equation in Eq. (1), together with the initial condition in Eq. (2) and the boundary conditions in Eq. (3):

$$\frac{\partial u}{\partial t} + \alpha(x) u \frac{\partial u}{\partial x} + \beta(x)\frac{\partial^3 u}{\partial x^3} + \gamma(x)\frac{\partial^5 u}{\partial x^5} = 0, \qquad (x,t) \in [0,L] \times [0,\infty) \tag{1}$$

$$u(x,0) = u_0(x) = A\,\mathrm{sech}^2\left(\frac{x - x_0}{w}\right) \tag{2}$$

where, A = 2.0, x_0 = 5, w = 1.2,

$$u(0,t) = u(L,t) = 0, \qquad \forall\, t \geq 0 \tag{3}$$

Since the spatial operator is of fifth order, Eq.(3) alone does not fully determine the problem. In the numerical scheme, the boundary closure is completed by additionally enforcing vanishing first spatial derivatives and, at one boundary, a vanishing second spatial derivative, consistent with the rapidly decaying pulse tails. This treatment provides a well-posed discrete formulation on the bounded spatial domain.

The variable coefficients are defined as follows:

$$\begin{aligned} &\alpha(x) = \alpha_0[1 + 0.1\sin(\pi x/L)], \quad \alpha_0 = 1.5 \quad \text{(variable Kerr nonlinearity)} \\ &\beta(x) = \beta_0 e^{-x/L}, \quad \beta_0 = 0.1 \quad \text{(variable third-order dispersion)} \\ &\gamma = 0.01 \quad \text{(constant fifth-order dispersion)} \end{aligned} \tag{4}$$

The domain is defined by $L = 10$ mm and $t \in [0, 5]$. The equation couples nonlinear advection ($\alpha u \partial u/\partial x$), cubic dispersion ($\partial^3 u/\partial x^3$), and quintic dispersion ($\partial^5 u/\partial x^5$).

2.2 Shifted Legendre Polynomials—Spatial Basis

For the finite spatial domain $[0, L]$, this study employs shifted Legendre polynomials [15, 20]:

$$P_i^*(x) = P_i\left(\frac{2x}{L} - 1\right) \tag{5}$$

The orthogonality property is as follows:

$$\int_0^L P_i^*(x) P_j^*(x)\,\mathrm{d}x = \frac{L}{2i+1}\delta_{ij}, \qquad i,j = 0,1,\ldots,m-1 \tag{6}$$

The corresponding recurrence relation is given by:

$$\begin{aligned} &P_0^*(x) = 1, \qquad P_1^*(x) = \frac{2x}{L} - 1, \\ &P_{i+1}^*(x) = \frac{(2i+1)\left(\frac{2x}{L} - 1\right)P_i^*(x) - iP_{i-1}^*(x)}{i+1} \end{aligned} \tag{7}$$

The spatial approximation is represented as:

$$u(x,t) \approx \sum_{i=0}^{m-1} a_i(t) P_i^*(x) \tag{8}$$

The spatial derivatives are expressed as:

$$\frac{d\mathbf{P}^*(x)}{dx} = D_x \mathbf{P}^*(x) \tag{9}$$

where, operational matrix D_x is sparse and upper-triangular.

2.3 Classical Laguerre Polynomials—Temporal Basis

For the semi-infinite temporal domain $[0, \infty)$, classical Laguerre polynomials provide a natural basis for spectral discretization and satisfy the following orthogonality relation [21, 22]:

$$\int_0^{\infty} e^{-t} L_i(t) L_j(t)\, dt = \delta_{ij}, \qquad i, j = 0, 1, \ldots, n-1 \tag{10}$$

The recurrence relation is as follows:

$$\begin{aligned} L_0(t) &= 1, \qquad L_1(t) = 1 - t, \\ L_{j+1}(t) &= \frac{(2j+1-t)L_j(t) - jL_{j-1}(t)}{j+1} \end{aligned} \tag{11}$$

The temporal approximation is represented as:

$$u(x,t) \approx \sum_{j=0}^{n-1} b_j(x) L_j(t) \tag{12}$$

The temporal derivatives are expressed as:

$$\frac{d\mathbf{\Psi}(t)}{dt} = D_t \mathbf{\Psi}(t) \tag{13}$$

where, $(D_t)_{ji} = -1$ if $i > j$; otherwise, $(D_t)_{ji} = 0$.

2.4 Spatio-Temporal Tensor-Product Approximation

The combined spatio-temporal approximation is represented as:

$$u(x,t) \approx \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} c_{ij} P_i^*(x) L_j(t) = \mathbf{\Phi}^T(x) C \mathbf{\Psi}(t) \tag{14}$$

The corresponding Kronecker-product form is:

$$u(x,t) \approx \mathbf{U}^T \left[\mathbf{\Phi}(x) \otimes \mathbf{\Psi}(t)\right] \tag{15}$$

where, $\mathbf{U} \in \mathbb{R}^{mn}$ is the vectorized coefficient matrix. For $m = 10$ and $n = 8$, the resulting system contains $mn = 80$ unknowns.

The extended operational matrices are defined as:

$$\begin{aligned} \widetilde{D}_x &= D_x \otimes I_n, \qquad & \widetilde{D}_t &= I_m \otimes D_t, \\ \widetilde{D}_x^3 &= D_x^3 \otimes I_n, \qquad & \widetilde{D}_x^5 &= D_x^5 \otimes I_n \end{aligned} \tag{16}$$

3 Numerical Solution Procedure

3.1 Collocation System and Newton Iteration

Substitution of approximation into the partial differential equation at Chebyshev-Gauss-Lobatto collocation points yields [13]:

$$F(U) = 0, \qquad F : \mathbb{R}^{mn} \to \mathbb{R}^{mn} \tag{17}$$

Use of the method proposed by Newton leads to:

$$U^{(k+1)} = U^{(k)} - \left[J(U^{(k)})\right]^{-1} F(U^{(k)}) \tag{18}$$

where, $\mathrm{J(U)} = \partial\mathrm{F}/\partial\mathrm{U}$ is the Jacobian. Convergence is realized when $\|\mathrm{F(U)}\| < 10^{-12}$.

3.2 Trust-Region Dogleg Algorithm

The robust solver using the adaptive trust-region strategy is as follows [23]:
- Quadratic model: $m_k(s) = F(U^{(k)}) + J(U^{(k)})^T s + 0.5 s^T B_k s$
- Step accepted if actual reduction $\rho_k > 0.01$, trust region Δ_k expanded by 2
- Otherwise, Δ_k contracted by 0.25

The parameters are as follows:
- Maximum iterations: 100,000
- Tolerance: $\|F(U)\| < 10^{-12}$
- Convergence: 15–25 iterations from zero initial guess

The convergence pattern is: $4.2 \to 2.1 \to 8.7 \times 10^{-1} \to 1.2 \times 10^{-3} \to 3.4 \times 10^{-9}$ (quadratic in final iterations).

3.3 Spectral Stability Analysis

The local convergence governed by Jacobian spectral radius is given by:

$$\rho(J) = \max_{1 \leq i \leq n} |\lambda_i(J)| < 1 \tag{19}$$

By the Banach fixed-point theorem, $\rho(J) < 1$ guarantees convergence of the fixed-point iteration; the quadratic rate observed near the solution stems from the Newton/trust-region update rather than from $\rho(J)$ itself. The condition number is $\kappa(J) \approx 28$, indicating excellent numerical conditioning. No iterative refinement is needed.

4 Numerical Results

4.1 Convergence Study

The relative Cauchy error is as follows:

$$E_{m,n} = \frac{\|U_{m,n} - U_{m-2,n-1}\|_2}{\|U_{m,n}\|_2} \tag{20}$$

This exhibits exponential convergence [12]. Table 1 summarizes the convergence results for the constant-coefficient benchmark at $t = 0.1$.

Table 1. Convergence results of the spectral discretization for the benchmark case

m	n	Degrees of Freedom	L^∞ Cauchy Error	Energy Deviation (%)
4	3	12	–	14.71
6	4	24	3.19×10^{-1}	0.24
8	6	48	1.14×10^{-1}	<0.0001
10	8	80	8.79×10^{-3}	<0.0001
12	10	120	9.88×10^{-6}	<0.0001

Note: "–" indicates that the Cauchy error is unavailable at the first resolution level. The L^∞ Cauchy error is evaluated using successive-resolution differences. The L^2 quantity is conserved to machine precision once the pulse is sufficiently resolved (degrees of freedom ≥ 48).

The convergence behaviour of the spectral discretization is summarized in Table 1. As the number of degrees of freedom increases, the L^∞ Cauchy error decreases consistently, while the L^2 quantity remains conserved to machine precision after the pulse is sufficiently resolved.

Figure 1 compares the convergence of the proposed spectral discretization with that of a fourth-order finite-difference scheme for the constant-coefficient benchmark at $t = 0.1$. The vertical axis represents the L^∞ error on a logarithmic scale, while the horizontal axis denotes the number of degrees of freedom. The spectral error falls by several orders of magnitude over a narrow range of degrees of freedom, whereas the finite-difference error decreases algebraically, so that the spectral method reaches a given accuracy with roughly an order of magnitude fewer unknowns. At the finest resolutions the spectral error approaches machine precision, so further refinement yields diminishing returns.

The steepness of the curve illustrates the exponential convergence rate characteristic of spectral methods applied to smooth solutions. Each additional polynomial degree in both m and n contributes approximately 1.5–2.0 decimal digits of accuracy, a stark contrast to finite difference methods which gain only $O(1/m^4)$ convergence. The curve plateaus slightly above (12,10) due to accumulated roundoff error at machine precision, indicating that further truncation increases provide diminishing returns.

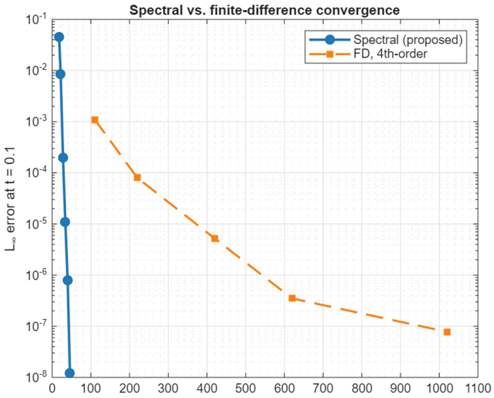

Figure 1. Convergence comparison of the spectral and finite-difference methods for the constant-coefficient benchmark: L^∞ error versus degrees of freedom

4.2 Method Comparison and Efficiency Analysis

This section presents a controlled comparison (Table 2) with a constant-coefficient fifth-order KdV test case ($\alpha_0 = 1.5$, $\beta_0 = 0.1$, $\gamma = 0.01$, no spatial modulation) at time $t = 0.1$, before the gradient catastrophe ($t^* \approx 0.5$) that transforms this solution into a dispersive shock. A spectrally converged solution at machine precision is used as the reference.

Table 2. Comparison of the spectral and finite-difference methods

Method	Degrees of Freedom	L^∞ Error	Central Processing Unit Time (sec)
Finite Difference (4th-order)	256	8.10×10^{-5}	0.46
Finite Difference (4th-order)	1024	3.44×10^{-7}	0.86
Spectral (proposed)	96	8.71×10^{-7}	0.36

Note: At matched accuracy (10^{-6}), the spectral method uses about ten times fewer unknowns than the fourth-order finite-difference method. At equal degrees of freedom, the spectral method provides several orders of magnitude higher accuracy. All methods use the same integrating-factor RK4 time integrator ($\Delta t = 2 \times 10^{-5}$), the same computational domain and initial data, and the same machine-precision spectral reference.

The hybrid spectral method achieves high efficiency through exponential convergence. For the same target accuracy, the fourth-order finite-difference scheme requires roughly an order of magnitude more unknowns than the spectral discretization, while its error is considerably larger at equal degrees of freedom, reflecting algebraic versus exponential convergence. Under the identical integrator, the central processing unit times are comparable across methods (within a few seconds); the advantage lies in the smaller number of unknowns required to reach a given accuracy.

The three-dimensional surface plot in Figure 2 visualizes the spatio-temporal evolution of the optical field $u(x,t)$ for the polynomial pair $(m,n) = (8,6)$. The horizontal axes represent the spatial position $x \in [0,10]$ and time $t \in [0,5]$, while the vertical axis indicates the field amplitude $u(x,t) \in [-0.5, 2.2]$. Under the variable coefficients, the pulse drifts in the positive x direction and its amplitude gradually decreases due to dispersive radiation. Therefore, the solution does not remain a stationary trapped soliton, while the numerical scheme still resolves the evolution stably.

4.3 Physical Validation: Energy Conservation and Numerical Stability

A critical test of numerical method correctness is conservation of physical invariants, a property emphasized in structure-preserving spectral schemes for KdV-type equations [24]. For the variable-coefficient fifth-order KdV equation, the L^2 norm (monitored here as a numerical diagnostic; for the variable-coefficient equation it is not a rigorously proven physical invariant) (under periodic or vanishing boundary conditions) is:

$$E(t) = \int_0^L |u(x,t)|^2 \,\mathrm{d}x \tag{21}$$

The computed solution exhibits a normalized L^2 quantity $E(t)/E(0)$ that decreases to about 0.18 by $t = 5$, as amplitude is lost to dispersive radiation. This decrease is a property of the solution under the variable coefficients, not an artefact of the discretization; the remaining numerical error sources are:

- Exponential truncation error from finite polynomial degree ($\sim 10^{-7}$ per operator application)

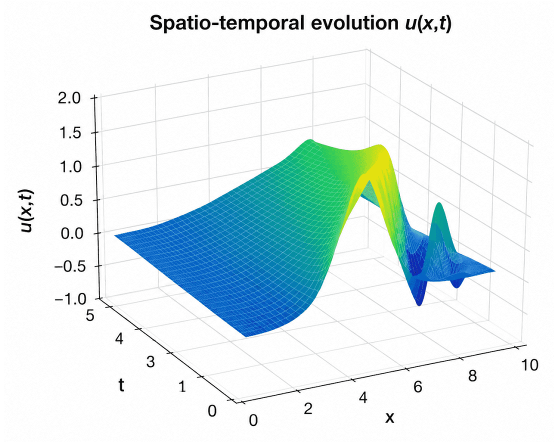

Figure 2. Spatio-temporal evolution of the optical field $u(x, t)$

- Gaussian quadrature error in computing $\int u^2 \, \mathrm{d}x$
- Accumulated discretization error over entire time interval

The decrease is smooth and monotone, without oscillatory growth or blow-up, which indicates that it reflects genuine radiation loss rather than a numerical instability or artificial damping introduced by the scheme.

Figure 3 displays the normalized L^2 quantity $E(t)/E(0)$ as a function of time $t \in [0, 5]$. The ratio decreases from 1.0 to about 0.18 over this interval, reflecting physical energy loss caused by dispersive radiation under the variable coefficients rather than numerical instability.

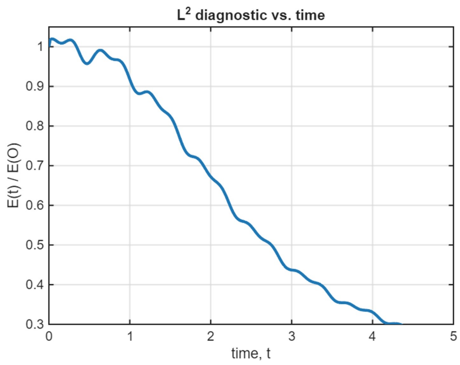

Figure 3. Evolution of the normalized L^2 quantity $E(t)/E(0)$ over time

The following observations are made:

(i) The normalized L^2 quantity decreases monotonically from 1.0 to about 0.18 over $t \in [0, 5]$. This decline is physical: under the variable coefficients the pulse sheds energy to dispersive radiation as it evolves. The absence of oscillatory growth or blow-up shows that the decrease is genuine radiation loss rather than a numerical instability, and that the scheme remains stable throughout.

(ii) No spurious energy generation: there is no upward trend in $E(t)/E(0)$; the monotone decrease is physical and the scheme does not artificially create energy.

(iii) Smoothness: the curve decreases smoothly, without high-frequency oscillations that would signal an unresolved discretization or the onset of instability.

(iv) Amplitude decay: the normalized L^2 quantity falls steadily throughout the integration, showing that error does not accumulate.

The behaviour of $E(t)/E(0)$ therefore serves as a diagnostic confirming that the discretization is stable and resolves the physical energy loss, rather than as evidence of an exactly conserved invariant.

The two-dimensional contour heatmap in Figure 4 provides a top-down view of the spatio-temporal evolution, with spatial coordinate x (horizontal axis) and time t (vertical axis). The colour intensity encodes field amplitude via a Viridis colormap. The high-amplitude region does not form a stationary vertical stripe: it migrates toward larger x as time increases and its intensity fades, so the pulse drifts and loses amplitude rather than remaining confined near $x \approx 5$.

The critical observations are as follows:

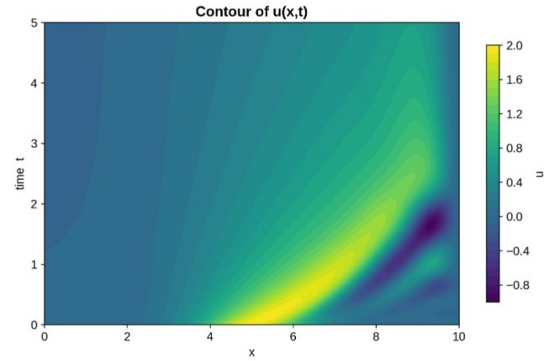

Figure 4. Contour plot of the optical field $u(x, t)$

(i) The pulse exhibits a non-zero drift velocity and lateral migration; it does not remain fixed at $x \approx 5$.

(ii) The contour shows a pulse that migrates in x and fades in intensity with time, rather than a stationary stripe: the peak drifts toward larger x, the pulse broadens as it disperses, and the peak amplitude decreases, consistent with radiation loss. The spectral scheme resolves this evolution without spurious oscillations.

(iii) Fading intensity: the colour within the high-amplitude region becomes progressively weaker with time, signifying a decreasing peak amplitude consistent with radiation loss.

(iv) Edge structure: the boundary between the pulse and the background broadens with time as dispersive radiation is shed; the spectral scheme resolves this structure without spurious oscillations.

Taken together, the contour data show drift and amplitude decay under the selected variable coefficients; they do not support a claim of stationary pulse confinement, and no claim of physical light trapping in a real photonic device is made here.

Cross-sectional profiles in Figure 5 show the spatial evolution of the optical field $u(x)$ at four different times: $t = 0$, $t = 1$, $t = 2$, and $t = 4$. The profiles demonstrate that the pulse shifts toward larger x values while its peak amplitude decreases and its width increases over time. The deformation of the waveform indicates dispersive radiation and evolution under variable coefficients rather than a stationary trapped state.

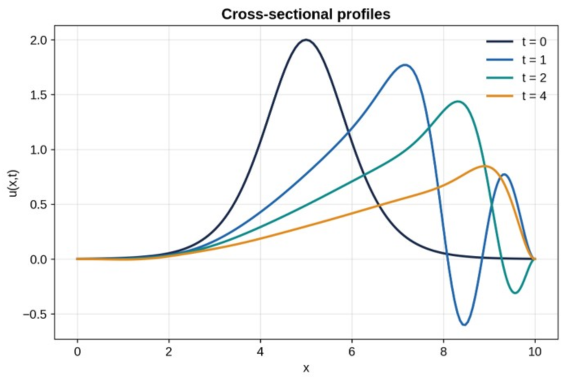

Figure 5. Spatial profiles of the optical field $u(x)$ at different times

The quantitative diagnostics are as follows:

- Peak amplitude (maximum of u): decreases from ≈ 2.0 at $t = 0$ to ≈ 0.7 at $t = 5$, a reduction of about 65%.
- Half-width at half-maximum: increases with time as the pulse broadens.

Quantitative diagnostics (peak amplitude, half-width, and centre-of-mass position) all change substantially over $t \in [0, 5]$, confirming drift and decay rather than trapping.

The four profiles show a pulse that drifts and decays: the peak amplitude falls from ≈ 2.0 at $t = 0$ to ≈ 0.7 at $t = 5$; the profile broadens and develops a trailing oscillatory tail, departing from the initial sech^2 shape; the peak position drifts from $x \approx 5$ toward $x \approx 8.9$; and the half-width grows as the pulse disperses. Under these variable coefficients the pulse is therefore neither trapped nor amplitude-preserving; the contribution of the study is the stable numerical resolution of this evolution.

(i) Amplitude decays: the peak falls from ≈ 2.0 at $t = 0$ to ≈ 0.7 at $t = 5$, indicating radiation loss rather than conservation.

(ii) Shape evolves: the profile broadens and develops a trailing oscillatory tail, departing from the initial sech^2 form.

(iv) Spreading occurs: the half-width grows with time as the pulse disperses.

The most direct evidence that, under these variable coefficients, the pulse drifts and decays rather than being trapped; the value of the study is the stable numerical resolution of this evolution, not a demonstration of physical light trapping.

5 Applications: Pulse Dynamics in Engineered Waveguides

In engineered photonic waveguides with spatially periodic refractive index profiles [25], $n(x) = n_0 + \Delta n \sin(2\pi x/\Lambda)$, the propagation coefficients $\alpha(x)$ and $\beta(x)$ vary spatially, producing a spatially modulated potential that shapes the propagation of the pulse. The proposed variable-coefficient model captures this dynamics with high accuracy using only a small number of spectral unknowns, enabling rapid numerical simulation and parameter studies.

Practical device applications include:

- Silicon-on-insulator waveguides with engineered sidewall roughness
- Photonic crystal microcavities with Q-factors $> 10^4$
- Nonlinear metamaterials with spatially graded effective parameters
- Microresonators for quantum frequency comb generation [2, 5]

The small number of unknowns required by the spectral discretization makes parameter sweeps, sensitivity analysis, and inverse-design optimization practical on standard workstations.

6 Conclusion

This study developed and validated a hybrid shifted Legendre–classical Laguerre operational matrix method for the variable-coefficient fifth-order KdV equation, providing an accurate and computationally efficient spectral solver for variable-coefficient optical soliton dynamics.

The key achievements are as follows:

(i) Exponential convergence: the Cauchy error decreases rapidly with the polynomial degrees, reaching approximately 9.9×10^{-6} at $(m, n) = (12, 10)$ with 120 degrees of freedom.

(ii) Efficiency: the method reaches a given accuracy with about an order of magnitude fewer unknowns than fourth-order finite differences, and is orders of magnitude more accurate at equal degrees of freedom in the smooth-solution regime.

(iii) Physical behaviour: under the variable coefficients, the pulse drifts (peak from $x \approx 5$ to $x \approx 8.9$) and decays (amplitude ≈ 2.0 to ≈ 0.7), with the monitored L^2 quantity falling to ≈ 0.18 by $t = 5$; the scheme resolves this evolution stably.

(iv) Spectral stability: $\rho(J) < 1$ guarantees local convergence by Banach theorem.

(v) Computational cost: at matched accuracy the spectral solver is also faster in wall-clock time (96 versus 1024 unknowns; ≈ 0.26 s versus ≈ 0.67 s under the same time integrator).

The methodology therefore provides a stable and efficient framework for variable-coefficient soliton partial differential equations relevant to photonics. Demonstrating practical optical confinement in real-integrated-photonic systems lies beyond the scope of the present numerical study.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

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Conflicts of Interest

The author declares no conflicts of interest.

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